

3D Embodied Agent-Mediated Affective Art Interaction for Emotion Externalization and Regulation

Xiang Li, Lei Meng*, Shiyu Zhu, Zongyi Yang, Yachong Zhang, Fangyuan Wei, Chunbo Li, and Xiangxu Meng

Abstract—Affective art interaction provides a low-threshold way for users to externalize and regulate their emotions through creative activities. Current intelligent affective art interaction systems predominantly rely on text- or audio-based guidance, successfully assisting users in expressing their emotions. However, these systems often fall short in providing sufficient social companionship, visual sensing, and embodied feedback throughout the creative process. To address this, we present a 3D embodied agent-mediated affective art interaction system. Rather than treating the agent as a general embodied conversational assistant, the proposed system centers on an Affective-Art Composition Game, where users express and reshape affect through interaction and agent-guided art design. The system features two main components: an Affective-Art Composition Game and an affective conversational agent with a customizable 3D avatar providing socially present and embodied feedback. To address insufficient implicit affect capture, we introduce a semantic-guided video emotion perception module that contributes facial-video evidence to user-state understanding. To improve the avatar’s behavioral feedback, we introduce a structure-aware avatar rigging module that supports stable embodied expression. On the MAFW benchmark, the proposed emotion-perception module achieves 44.45% UAR and 58.67% WAR, while the rigging module reduces joint-localization and skinning errors. User evaluations indicate that enhanced sensing improves perceived empathy and that improved rigging strengthens social presence and affinity. Collectively, this paper shows the potential of 3D embodied agent-mediated affective art interaction.

Index Terms—Affective intelligence, affective art interaction, emotion externalization, emotional well-being, 3D avatar, embodied feedback, video emotion perception.

I. INTRODUCTION

AFFECTIVE intelligence is increasingly expected to support interactive experiences in which users can express, inspect, and reshape their affective states. Recent affective conversational agents have shown the potential of always-available emotional support through dialogue, personalized responses, and strategy-aware response generation [1]–[4]. However, emotional experience is often ambiguous, private, and difficult to verbalize. Language-centered systems may therefore miss affective evidence expressed through nonverbal behavior, creative actions, and interaction patterns.

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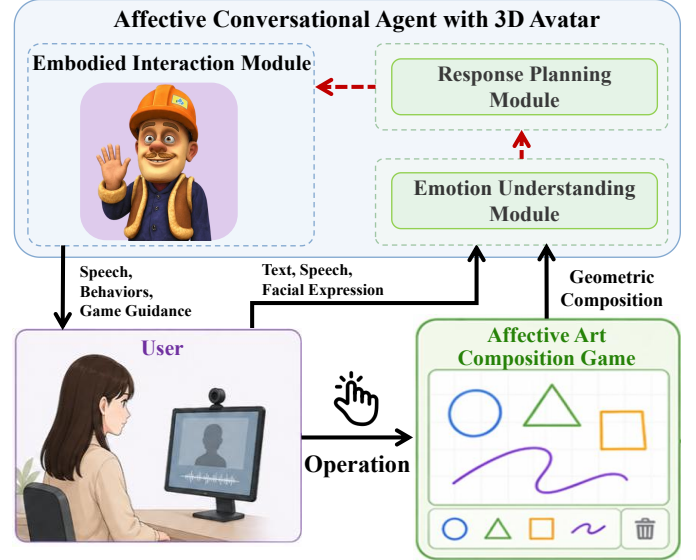


Fig. 1. 3D Embodied Agent-Mediated Affective Art Interaction framework. The user externalizes affect through the Affective-Art Composition Game, while the 3D embodied agent provides companionship, affective feedback, and game guidance based on speech/text context, facial video observations, and geometric composition.

Artistic interaction provides a promising medium for addressing this limitation. In this paper, we use *affective art* to refer to creative interaction that supports emotional expression, reflection, and regulation beyond artistic production alone. We use *emotion externalization* to describe the process of transforming internal affective states into visible, manipulable, and discussable external forms. Prior studies in art therapy and visual-affective design suggest that visual expression, geometric form, color, and composition can carry affective meaning and support emotional externalization [5]–[9]. This makes affective art interaction suitable for non-clinical emotional-support scenarios, where the goal is accessible expression, companionship, and short-term affective support rather than diagnosis, therapy, or treatment.

Despite these advantages, affective art interaction has long been supported mainly through relatively traditional forms of facilitation. Early art-therapy and visual-art practices often rely on therapist-guided sessions, final artwork interpretation, or pre/post-session outcome assessment [7]–[12]. In such settings, users can externalize emotions through visual creation, but the ongoing creative process is usually not computationally

perceived and interpreted. More recent digital and AI-assisted methods have improved the accessibility of art-based expression through digital media, generative support, and human–AI co-creation [5], [6], [13]–[20]. However, these intelligent supports are still often organized around text, speech, or prompt-level interaction, and they provide limited mechanisms for users’ visual emotional perception and enhancing the sense of companionship in the interaction. Recent affect-recognition research provides the perception side of embodied interaction: in-the-wild video datasets and multimodal models support facial-video and broader multimodal emotion understanding [21]–[24]. Separately, avatar-based and embodied-agent studies suggest that visual embodiment, emotion-aware voice, and behavioral feedback can strengthen social presence, affinity, or perceived companionship [25]–[31]. However, existing systems often treat creative expression, affect sensing, and embodied feedback as separate components, rarely leveraging a 3D embodied agent to connect art expression with how the system perceives, understands, responds, and facilitates adaptive emotional regulation.

To address this challenge, we propose a 3D embodied agent-mediated affective art interaction system as shown in Figure 1. The central idea is not to build a general embodied conversational agent, but to use the 3D embodied agent as a socially present companion that transforms affective art interaction from one-way user creation into a continuous process of creation, perception, embodied companionship, affective feedback, and agent-guided art design. The system consists of an Affective-Art Composition Game, where users externalize and reconstruct emotions through geometric visual composition, and a 3D avatar-based affective conversational agent, which provides verbal, vocal, behavioral, and game-guidance feedback based on speech/text context, facial video observations, and affective art interaction states. To support personalized embodied companionship, the agent also allows users to customize a 3D avatar from text or image prompts and then makes the generated avatar motion-ready through automatic skeleton rigging and skinning. To address insufficient implicit affect capture, we develop a semantic-guided video emotion perception module that extracts local facial semantic tokens and models temporal expression dynamics for user-state understanding. To improve embodied behavioral feedback for generated or stylized avatars, we develop a structure-aware avatar rigging module that improves skeleton localization and skinning consistency. These modules are designed to serve affect perception, avatar customization, and stable 3D embodied expression within the interaction system.

We evaluate the system through module-level experiments and a user study. On the MAFW benchmark, the semantic-guided video emotion perception module achieves 44.45% UAR and 58.67% WAR. The structure-aware avatar rigging module reduces skeleton localization and skinning errors under standard and pose-augmented character evaluation conditions. We further conduct a 2×2 within-subject user study with 25 participants to examine how affect sensing capability and avatar feedback quality influence user experience. The results show that stronger affect sensing mainly improves perceived empathy, stronger avatar rigging mainly improves social pres-

ence and affinity, and the combined condition produces the best task experience and the most favorable short-term self-reported affective changes.

The main contributions of this paper are summarized as follows:

- We propose a 3D embodied agent-mediated affective art interaction framework that organizes emotion externalization, user-state perception, embodied feedback and emotional regulation.
- We develop a 3D embodied affective conversational agent that supports text/image-based avatar customization and integrates speech/text context, facial-video evidence, and affective art interaction states to provide coordinated verbal, vocal, behavioral, and game-guidance feedback, supported by the proposed semantic-guided video emotion perception and structure-aware avatar rigging module.
- We conduct module-level experiments and a user study, which show that affect sensing and 3D embodied feedback improve perceived empathy, social presence, task experience, and short-term self-reported affective change in non-clinical affective art interaction.

II. RELATED WORK

A. Affective Art and Emotion Externalization

Affective art provides a nonverbal channel for expressing and reflecting on emotions through color, shape, contour, layout, and composition [32]–[34]. Guided drawing, coloring, and mindfulness-based visual creation have been associated with improvements in anxiety, stress, or negative affect for some populations and outcomes [7]–[12]. The evidence remains heterogeneous: a recent meta-analysis reported low overall study quality and improvement in only a minority of outcomes, precluding claims of stable or general therapeutic efficacy [9]. Digital and conversational systems can make visual expression more accessible [5], [6], [13]–[18].

Existing work nevertheless emphasizes final artifacts, therapist-guided sessions, initial expression, or post-session outcomes. Guided art-making agents and art-therapy chatbots remain mainly voice- or text-based [15], [16], and rarely treat the evolving artwork as user-state evidence for subsequent guidance. Our game-mediated composition instead uses creations, edits, and reconstruction behavior as contextual evidence within an emotion-externalization and affective-support process.

Conceptual positioning. Traditional art therapy is therapist- or clinician-led and may pursue assessment or treatment, whereas our prototype is non-clinical and neither diagnoses users nor assigns fixed psychological meanings to artwork [8], [10], [11]. AI-assisted art systems primarily support artifact generation or co-creation; here, users construct and revise the composition themselves, and canvas states and actions serve only as contextual evidence [14], [17], [18]. General emotional-support chatbots are language-centered, whereas our system couples the evolving creative state with facial-video evidence and coordinated verbal, vocal, avatar, and optional game guidance [1]–[3]. The contribution is this integrated affective-art interaction loop, not a replacement for therapy,

an autonomous art generator, or a general-purpose support chatbot.

B. Multimodal Emotion Recognition

Multimodal emotion recognition combines language, voice, facial behavior, and task context. Text methods model utterance-level emotion and conversational roles [35]–[37], while speech methods capture prosody and temporal structure [38], [39]. Cross-corpus speech recognition further reduces corpus-dependent acoustic variation through invariant feature learning and hierarchical or low-rank joint distribution adaptation [40]–[42]. Physiological emotion recognition also uses neural affective signals, including EEG-based emotional lateralization and spatiotemporal modeling, cross-subject domain adaptation, and subjective–objective feature fusion [43]–[45]. These cues remain complementary rather than individually definitive.

Visual emotion understanding spans image and video evidence [46], [47]. Dynamic facial-expression methods model temporal evolution on in-the-wild datasets including DFEW, FERV39k, and MAFW [21]–[23], [48], using expression intensity, self-supervised temporal learning, audio-visual relations, and fine-grained alignment [24], [49]–[58]. Sentiment analysis and MLLM prompting likewise combine semantic, visual, and contextual cues [59]–[61].

Most methods are still evaluated as offline classifiers that output labels or probabilities for isolated inputs. Interactive support instead requires facial, linguistic, vocal, task, and action evidence to be interpreted together. Although multimodal dialogue systems incorporate visual context [62], [63], they rarely connect affect cues with evolving creative states. We therefore use speech/text, facial video, and art-interaction state as complementary evidence rather than deterministic labels.

C. Affective Conversational and Embodied Agents

Affective conversational agents model supportive strategies, empathetic responses, and flow-aware emotional support language modeling [1]–[4], [64]. Speech and embodied agents add prosody, expression, gaze, posture, and motion to improve social presence [25]–[31]; 3D talking-head work also disentangles emotional facial motion for controllable animation [65].

Most agents remain language-centered, with speech, vision, avatar behavior, and creative activity only loosely coupled. Affective art interaction instead requires creative externalization, multimodal sensing, embodied feedback, guided reconstruction, and affective support to operate as one process.

III. 3D EMBODIED AGENT-MEDIATED AFFECTIVE ART INTERACTION

Figure 2 shows the overall framework of the proposed 3D embodied agent-mediated affective art interaction system. The system contains four modules: The first module is the Affective-Art Composition Game, which enables users to externalize affect through simple geometric composition and records the evolving canvas state and operation history. The second module is semantic-guided video emotion perception,

which extracts facial-video evidence to support implicit user-state understanding during interaction. The third module is 3D avatar customization and rigging, which generates personalized avatar appearances and prepares stable, motion-ready avatars for embodied behavioral feedback. The fourth module is state-aware response planning and generation, which integrates speech/text context, facial-video observations, and affective art interaction states to coordinate the agent’s verbal response, affective speech delivery, avatar behavior, and game-guidance feedback.

A. Design of Affective-Art Composition Game

1) *Game-Mediated Emotion Externalization:* The Affective-Art Composition Game is a low-threshold medium for emotion externalization and reconstruction. It uses simple geometric primitives, such as points, lines, circles, rectangles, triangles, rounded rectangles, and color blocks, which users can drag, rotate, scale, copy, delete, align, and rearrange.

A game session includes preparation, initial externalization, free composition, guided reconstruction, and reflective summary. The avatar provides a gentle prompt, optional guidance, and a short supportive reflection. The system does not judge the artwork or assign clinical meanings to it. Instead, the visual composition provides an external object that the user and the avatar can jointly inspect, discuss, and reshape.

2) *Game-State Encoding:* Let C_t , O_t , and R_t denote the canvas state, recent operation sequence, and reconstruction trajectory, respectively. The interpretable game state is

$$w_t = \phi_g(C_t, O_t, R_t) = [q_t^c, q_t^o, q_t^r], \quad q_t^r = q_t^c - q_0^c. \quad (1)$$

The descriptors cover shape and color distribution, density, sharpness, blank space, balance, repetition, edit type, idle time, undo frequency, and accepted guidance. They are contextual interaction cues rather than fixed psychological labels. Full definitions and extraction procedures are provided in Supplementary Material, Sec. S1.

3) *Game-Guidance Generation:* Based on game-state evidence, speech/text context, and video-based affective evidence, the system generates lightweight and optional guidance:

$$g_t = \Gamma_g(w_t, p_t^v, u_t), \quad (2)$$

where p_t^v denotes video-based emotion evidence and u_t denotes speech/text context. Guidance may suggest softening sharp elements, increasing blank space, rebalancing crowded regions, or focusing on a local change, and is presented as encouragement rather than correction.

B. Semantic-Guided Video Emotion Perception

1) *Semantic Facial Region Tokenization:* The video perception module provides implicit affective evidence for the interaction system. Inspired by the Facial Action Coding System [66], we extract local semantic tokens from facial regions. Given a facial video clip with T frames, Grounded SAM 2, which combines the Grounded SAM pipeline with SAM 2 video segmentation [67], [68], is used to obtain semantic facial region masks:

$$M \in \mathbb{R}^{T \times K \times H \times W}, \quad (3)$$

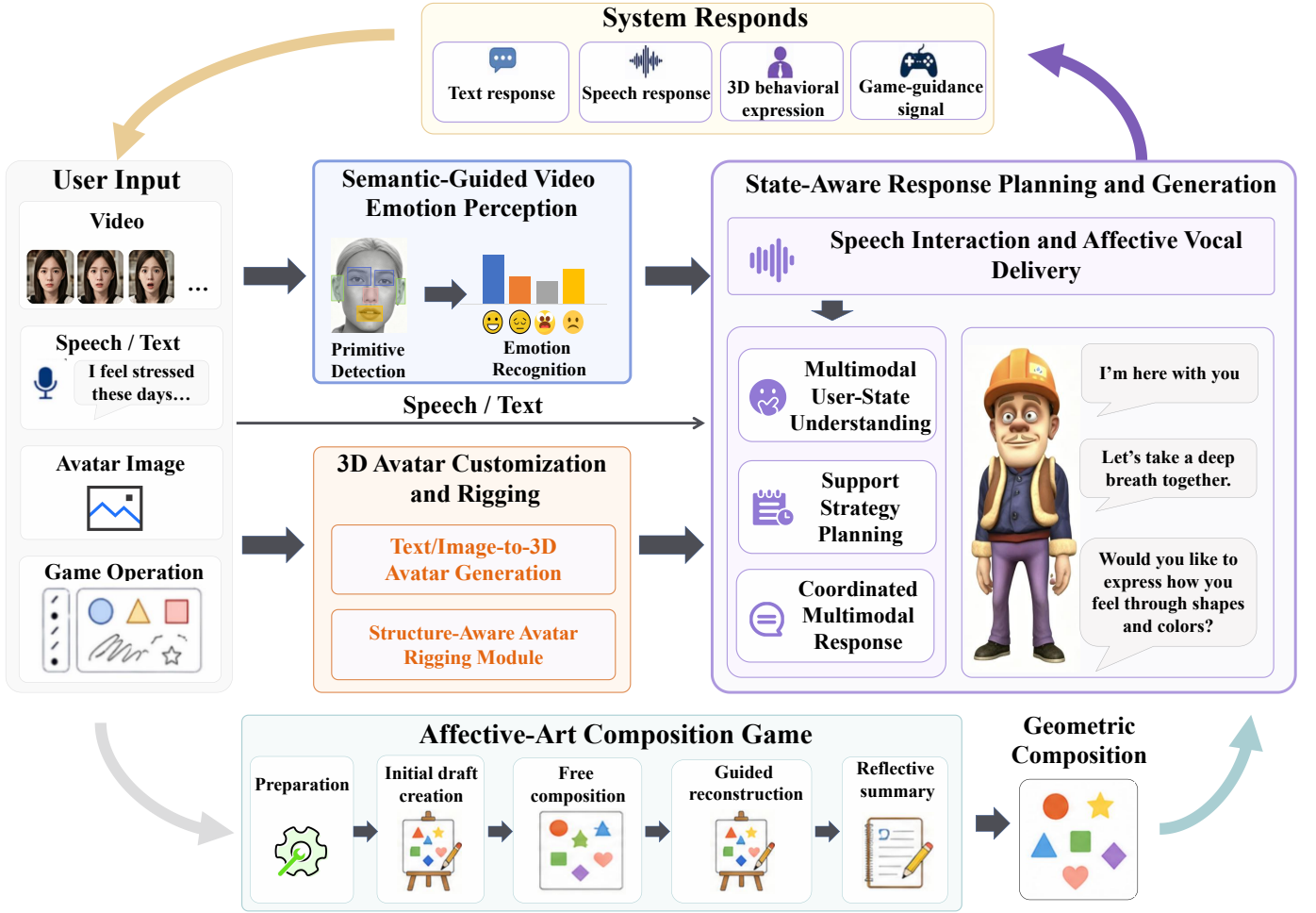


Fig. 2. Overview of the proposed 3D embodied agent-mediated affective art interaction system. The user externalizes affect and receives short-term support through the dialogue and Affective-Art Composition Game, while speech/text context, facial video observations, and affective art interaction states are integrated into the 3D embodied agent for user-state understanding and response generation.

where K denotes the number of semantic facial regions. Let F_t denote the frame-level feature map of the t -th frame. The token of the k -th region is computed as:

$$z_{t,k} = \text{Proj}(\text{AvgPool}(F_t \odot M_{t,k}) \parallel \text{MaxPool}(F_t \odot M_{t,k})), \quad (4)$$

where $\odot$ denotes masked selection and $\parallel$ denotes channel-wise concatenation.

2) *Spatiotemporal Semantic-Global Interaction*: To capture temporal expression dynamics, we compute the region-level motion token:

$$\Delta z_{t,k} = z_{t,k} - z_{t-1,k}. \quad (5)$$

The semantic tokens and their temporal differences are encoded by a temporal encoder:

$$\tilde{z}_k = \mathcal{T}_v([z_{1,k}, \dots, z_{T,k}, \Delta z_{2,k}, \dots, \Delta z_{T,k}]). \quad (6)$$

Let $\tilde{Z}$ denote the set of regional semantic features and G denote the global visual tokens from the video backbone. We perform semantic-global interaction as:

$$Z' = \tilde{Z} + \text{MHCA}(\tilde{Z}, G, G), \quad (7)$$

$$Z'' = Z' + \text{MHSA}(Z'), \quad (8)$$

$$G' = G + \text{MHCA}(G, Z'', Z''), \quad (9)$$

where $\text{MHCA}(\cdot)$ and $\text{MHSA}(\cdot)$ denote multi-head cross-attention and self-attention, respectively.

3) *Semantic-Guided MoE*: Following the basic setting of S4D [54], we use a ViT-B/16 backbone with a sparse mixture-of-experts layer (MoE). The expert-routing weights are conditioned on the pooled semantic representation:

$$\bar{z} = \text{Pool}(Z''), \quad \alpha = \text{Softmax}(\text{TopK}(W_g \bar{z} + \epsilon)). \quad (10)$$

The routed global representation is fused with the semantic representation:

$$f_t^v = \text{Concat}(G_{\text{out}}, \bar{z}), \quad (11)$$

and the video-based emotion-evidence distribution is:

$$p_t^v = \text{Softmax}(W_c f_t^v + b_c). \quad (12)$$

The module is trained with cross-entropy loss:

$$\mathcal{L}_v = - \sum_c y_c \log p_{t,c}^v. \quad (13)$$

In the proposed system, f_t^v and p_t^v are used as affective evidence rather than standalone diagnostic conclusions.

C. 3D Avatar Customization and Rigging

1) *Customization and Rigging Objective*: The 3D avatar provides the embodied feedback channel of the system. Its appearance can be generated by an existing text/image-to-3D generation tool, such as Hunyuan3D [69]. The focus of this work is to make the generated avatar motion-ready for supportive behaviors, such as nodding, leaning forward, opening the hands, pointing to game regions, and adjusting posture.

Given an avatar mesh sampled as a point set $P \in \mathbb{R}^{N \times 3}$, the rigging module estimates a skeleton $J \in \mathbb{R}^{K \times 3}$ and a skinning-weight matrix $A \in \mathbb{R}^{N \times K}$:

$$J, A = \mathcal{R}(P). \quad (14)$$

Accurate joint localization and coherent skinning are important for avoiding distorted limbs, unstable gestures, and mismatched body movements during embodied affective feedback.

2) *Structure-Aware Avatar Rigging Module*: The module is built on a UniRig-style automatic rigging pipeline [70]. A base rigging network first predicts initial joints and skinning weights:

$$J^0, A^0 = \mathcal{R}_0(P). \quad (15)$$

To explicitly connect surface geometry with skeleton hierarchy, let $H = [h_1, \dots, h_K]$ denote the skeleton hierarchy embedding. For each surface point p_i , the structure-aware binding feature is:

$$b_i = \left[p_i, \sum_{k=1}^K A_{ik}^0 h_k \right]. \quad (16)$$

The complete avatar representation is:

$$B = \{b_i\}_{i=1}^N. \quad (17)$$

A refinement network predicts updated joints and skinning weights:

$$J^r, A^r = \mathcal{R}_r(B). \quad (18)$$

To reduce skeleton-skinning inconsistency, a skinning-derived skeleton estimate is computed as:

$$J_k^s = \frac{\sum_{i=1}^N A_{ik}^r p_i}{\sum_{i=1}^N A_{ik}^r + \epsilon}. \quad (19)$$

The final skeleton is obtained by fusing the initial, refined, and skinning-derived estimates:

$$J = J^r + \mathcal{F}([J^0, J^r, J^s]), \quad (20)$$

where $\mathcal{F}(\cdot)$ predicts a residual skeleton displacement. The final rig is represented as (J, A^r) .

When supervised rigging annotations are available, the module is optimized by:

$$\mathcal{L}_{rig} = \lambda_J \|J - J^*\|_1 + \lambda_A \|A^r - A^*\|_1 + \lambda_S \sum_{k=1}^K \|J_k - J_k^s\|_1, \quad (21)$$

where J^* and A^* denote the ground-truth skeleton and skinning weights.

3) *Embodied Behavior Response*: The rigged avatar is used as the behavioral embodiment of the response plan. We pre-build a small action library based on Mixamo motions and retarget these motions to the rigged avatar for supportive behaviors. For example, validation can be paired with steady gaze and slight nodding, reassurance with relaxed posture, grounding with slow breathing-like motion, and creative guidance with pointing gestures toward game regions.

D. State-Aware Response Planning and Generation

1) *Speech Interaction and Affective Vocal Delivery*: The speech interface connects user speech input and agent vocal output within the response-planning pipeline. User speech is transcribed by Fun-ASR [71] and used as part of the speech/text context for multimodal user-state understanding. For agent output, the verbal response is synthesized through an external Fun-CosyVoice TTS service [72]. Although the speech synthesis model is implemented using an external TTS service, the proposed system explicitly models affective speech delivery as part of the response plan. The speech-delivery intent controls pace, pauses, emphasis, and affective vocal style, ensuring that the verbal response and avatar behavior are expressed consistently.

2) *Multimodal User-State Understanding*: The agent does not convert multimodal inputs into a deterministic emotion label. Each turn is summarized by recent speech/text, video-based emotion evidence and confidence, game-state cues, and interaction status. Direct self-reports and semantically explicit speech/text are primary; video evidence is weighted by confidence, and game state is contextual only. If sources disagree, the summary preserves the conflict. The planner then uses a neutral reflection or clarifying question and a low-risk, non-clinical strategy, and inferred cues never override an explicit user statement.

3) *Support Strategy Planning*: The planner maps the state summary to a constrained non-clinical strategy and coordinated verbal, vocal, avatar, and optional game-guidance intents. For example, if a user reports pressure while moderate-confidence video and repeated undo or idle behavior provide tentative contextual support, the summary keeps the direct report primary and selects reassurance with grounding. The response acknowledges pressure without assigning an emotion label, slows its pace, uses an open posture and slight nod, and may suggest creating a small area of blank space. A complete state-to-response walkthrough is provided in Supplementary Material, Sec. S2.

As methodological inspiration rather than direct evidence for affective-support effectiveness, tree-structured reasoning in knowledge-graph question answering illustrates the use of explicit intermediate structures for faithful generation [73]. Following this design principle, the planner exposes state summaries and response intents before final multimodal response generation.

The predefined strategy vocabulary adapts validation, reassurance, reflection, and encouragement from emotional-support dialogue taxonomies [1]–[3], [64], while grounding and creative guidance are low-risk and task-specific heuristics. The prompt-constrained planner is a non-clinical support

policy, not a learned therapeutic policy; full provenance is provided in Supplementary Material, Sec. S2.

4) *Coordinated Multimodal Response Generation*: The final response is produced as a coordinated multi-channel action. The verbal response provides validation, reflection, encouragement, or optional art-making suggestions; the speech-delivery control specifies pace, pauses, emphasis, and affective vocal style; the avatar behavior maps the strategy to gaze, posture, gesture, and timing; and the game guidance translates the same strategy into lightweight creative operations, such as softening a crowded region, adding blank space, adjusting color contrast, or focusing on a small local change.

5) *Module Reliability and Fallbacks*: Availability and confidence flags permit graceful degradation: invalid modalities are excluded, unavailable output channels are skipped, and neutral text remains available without fabricating evidence or deterministic emotion labels. The module-specific fallback matrix is provided in Supplementary Material, Sec. S2.

IV. VIDEO EMOTION PERCEPTION EVALUATION

A. Experimental Setting

Following standard practices in dynamic facial expression recognition, we employ Unweighted Average Recall (UAR) and Weighted Average Recall (WAR) as the primary evaluation metrics to account for class imbalance. To leverage generalized large-scale visual representations, we initialize our model using the S4D checkpoint [54], which was pre-trained on the VoxCeleb2 [74] and AffectNet [75] datasets. The model is subsequently fine-tuned on the challenging in-the-wild MAFW [23] dataset.

All baseline results in Table I were reproduced by us using the official GitHub implementations and released configurations, rather than copied from published tables. Every method used the same official MAFW fold partitions and common UAR/WAR evaluation. Model-specific preprocessing and input configurations, including required modalities and temporal sampling, were retained from the official implementations.

B. Compared Methods

We compare our proposed semantic-guided video emotion perception module against several recent state-of-the-art Dynamic Facial Expression Recognition (DFER) methods. The baselines cover a range of architectures and modalities, including self-supervised video learning approaches such as VideoMAE [49], MAE-DFER [51], and HiCMAE [52]; the multimodal contrastive learning framework FineCLIPER [53]; and recent advanced models like LGTC-DFER [55] and S4D [54]. The common MAFW partitions and evaluation code provide a consistent benchmark, while official model-specific preprocessing preserves each baseline's intended implementation.

C. Main Results

Table I presents the quantitative comparison of our model against the baselines on the MAFW dataset. Our proposed

model achieves 44.45% UAR and the highest WAR of 58.67% among all compared methods.

Relative to reproduced video-only S4D, our method improves UAR/WAR by 1.51/1.06 points; the gains over video-only LGTC-DFER are 0.34/0.51 points. FineCLIPER with video and text has 0.56-point higher UAR, whereas our video-only model has 1.76-point higher WAR. These are consistent but moderate gains; the ablation study provides more direct evidence for the proposed components.

D. Ablation Study

Table II summarizes our ablation study on the MAFW dataset. First, removing the **Spatiotemporal Semantic-Global Interaction** drops performance to 40.66% UAR, confirming that local facial tokens require explicit semantic-global cross-attention to avoid isolated, noisy representations. Second, replacing the **Semantic-Guided MoE** with standard MoE routing yields 43.82% UAR; while this validates the interaction module alone, achieving our peak performance (44.45% UAR) requires conditioning the expert routing weights (α) on the pooled semantic representation ($\bar{z}$) to effectively capture fine-grained expression dynamics.

V. 3D AVATAR RIGGING EVALUATION

A. Experimental Setting

We conduct experiments on Mixamo and VRoid characters [76], [77], as well as their pose-augmented variants, Mixamo+ and VRoid+. Mixamo contains humanoid characters with relatively regular skeletons, while VRoid contains stylized characters with different body proportions and surface details. The augmented variants introduce more challenging poses and geometry variations, making them closer to generated or retargeted avatars used in our system. Following common automatic-rigging evaluation practice, we report three metrics: CD-J2J for joint-to-joint skeleton localization error, CD-B2B for bone-to-bone structural discrepancy, and Skin-L1 for skinning-weight error. Lower values indicate better performance for all metrics. We compare our method with RigNet [78] and UniRig [70].

B. Main Results

As shown in Table III and Figure 3, the proposed module achieves the best or tied-best performance across all four evaluation conditions and consistently reduces skeleton-related errors relative to UniRig, especially in pose-augmented settings. More accurate skeletons and coherent skinning reduce distorted limbs and unstable supportive gestures.

The four evaluation conditions—Mixamo, VRoid, and their pose-augmented variants—cover regular and stylized humanoids with some pose and geometry variation; they do not establish reliability for non-humanoids, extreme proportions, or arbitrary generated topology. Robustness claims are therefore limited to the evaluated humanoid settings.

TABLE I
COMPARISON WITH RECENT STATE-OF-THE-ART DFER BASELINES ON THE MAFW DATASET.

Method	Backbone	Modality	UAR(%)	WAR(%)
VideoMAE [49]	ViT-B/16	Video	40.87	53.51
MAE-DFER [51]	ViT-B/16	Video	41.62	54.31
HiCMAE [52]	ViT-B/16	Video	42.10	54.84
HiCMAE [52]	ViT-B/16	Video + Audio	42.65	56.17
FineCLIPER [53]	CLIP-ViT-B/16	Video	43.02	54.69
FineCLIPER [53]	CLIP-ViT-B/16	Video + Text	45.01	56.91
LGTC-DFER [55]	ViT-B/16	Video	44.11	58.16
S4D [54]	ViT-B/16	Video	42.94	57.61
S4D [54]	ViT-B/16	Video + Image	43.72	58.44
Our Model	ViT-B/16	Video	44.45	58.67

TABLE II
ABLATION STUDY OF THE SEMANTIC-GUIDED VIDEO EMOTION PERCEPTION MODULE.

Method	UAR(%)	WAR(%)
Baseline (S4D: Global ViT + MoE)	42.94	57.61
w/o Spatiotemporal Semantic-Global Interaction	40.66	55.90
w/o Semantic-Guided MoE	43.82	58.38
Our Full Model	44.45	58.67

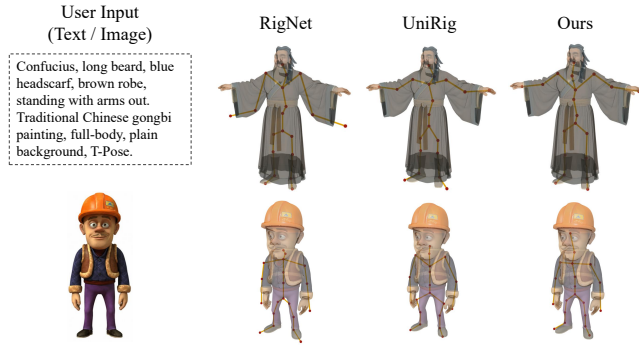


Fig. 3. Qualitative comparison of avatar rigging results. The proposed module produces skeletons that are more consistent with avatar geometry, supporting stable motion retargeting.

C. Ablation Study

Table IV reports the component ablation. SAE improves geometry–skeleton relations, while MSF refines joint placement by combining initial, refined, and skinning-derived skeletons. The full model gives the strongest overall skeleton and skinning accuracy, showing that the two components provide complementary gains.

VI. USER STUDY OF EMBODIED AFFECTIVE ART INTERACTION

A. Prototype Implementation

We implemented a prototype system to evaluate whether a 3D embodied agent with affect sensing, state understanding, and behavioral feedback can improve affective art interaction. The front end uses React, Capacitor, and Three.js to render the 3D avatar and the Affective-Art Composition Game. The backend coordinates speech recognition, dialogue generation, speech synthesis, avatar animation, and game-state logging,

integrating Fun-ASR [71], Fun-CosyVoice [72], and Qwen3-VL-Flash through the DashScope API. The avatar interface was built on the Fay digital-human framework [79] and extended with our structure-aware rigging pipeline for behavior retargeting. The prototype was also deployed and tested on an Android-based companion display device to evaluate its feasibility in an integrated interactive setting. Figure 4 shows the prototype interface.

Component provenance. The prototype integrates cited models, services, frameworks, and motion assets; our contributions are the game-state interaction, semantic-guided video module, structure-aware rigging components, and evidence-to-response coordination. We do not claim upstream components as original; the complete component-to-contribution map is provided in Supplementary Material, Sec. S2.

The video module ran on a single NVIDIA RTX 4090 GPU. Each update uniformly samples 16 frames from a 4-s video window for Grounded SAM 2 facial-region extraction and emotion inference. The 4-s value is the input duration, not computation latency. Processing delay was small relative to this window, so the evidence was available for the next planning update. We do not claim separately benchmarked real-time latency; delay may vary with hardware and service load.

B. Objectives and Hypotheses

The user study evaluates whether the proposed 3D embodied agent improves affective art interaction and emotional support in a non-clinical setting. Unlike the preceding technical experiments, this study examines user-perceived benefits in emotional understanding, embodied companionship, task experience, emotion externalization, and immediate self-reported affective change. We examine two agent capabilities through a 2×2 within-subject design: video-based affect sensing for perceiving and analyzing the user’s affective state, and avatar rigging quality for stable, natural 3D behavioral feedback. This design allows us to evaluate whether stronger sensing and stronger embodied feedback independently and jointly improve the affective art interaction experience.

Based on these objectives, we formulate three hypotheses:

- **H1:** Stronger affect sensing and state understanding improve perceived emotional understanding, empathy, and response quality during affective art interaction.

TABLE III
COMPARISON WITH RIGGING BASELINES UNDER STANDARD AND POSE-AUGMENTED CHARACTER EVALUATION CONDITIONS. LOWER IS BETTER FOR ALL METRICS.

Method	Mixamo			VRoid			Mixamo+			VRoid+		
	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1
RigNet [78]	0.1022	0.0205	0.0908	0.2405	0.0676	0.0489	0.2484	0.0605	0.1073	0.2171	0.0714	0.1292
UniRig [70]	0.0116	0.0069	0.0165	0.0044	0.0034	0.0047	0.0298	0.0221	0.0335	0.0161	0.0115	0.0288
Ours	0.0085	0.0049	0.0152	0.0034	0.0021	0.0047	0.0236	0.0171	0.0304	0.0090	0.0067	0.0251

TABLE IV
ABLATION OF STRUCTURE-AWARE AVATAR ENCODING (SAE) AND MULTI-SKELETON CONSISTENCY FUSION (MSF). LOWER IS BETTER.

Method	Mixamo			VRoid			Mixamo+			VRoid+		
	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1	CD-J2J	CD-B2B	Skin-L1
Baseline (UniRig)	0.0116	0.0069	0.0165	0.0044	0.0034	0.0047	0.0298	0.0221	0.0335	0.0161	0.0115	0.0288
w/o SAE	0.0097	0.0053	0.0160	0.0035	0.0021	0.0047	0.0262	0.0196	0.0324	0.0145	0.0107	0.0300
w/o MSF	0.0107	0.0064	0.0152	0.0044	0.0034	0.0047	0.0253	0.0187	0.0304	0.0110	0.0087	0.0251
Full	0.0085	0.0049	0.0152	0.0034	0.0021	0.0047	0.0236	0.0171	0.0304	0.0090	0.0067	0.0251

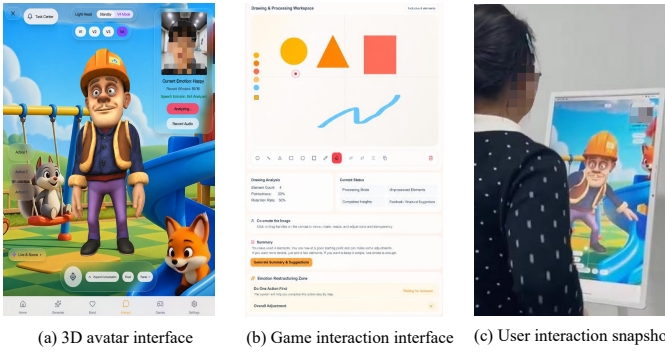


Fig. 4. Prototype interface of the proposed system. The figure shows the 3D avatar interaction interface and the Affective-Art Composition Game interface.

TABLE V
EXPERIMENTAL CONDITIONS IN THE 2×2 WITHIN-SUBJECT USER STUDY.

Condition	Affect sensing capability	Avatar rigging capability
C1	Weak	Weak
C2	Strong	Weak
C3	Weak	Strong
C4	Strong	Strong

- **H2:** Stronger 3D behavioral feedback improves perceived social presence, avatar affinity, and embodied feedback quality.
- **H3:** A 3D embodied agent with both stronger affect sensing and stronger behavioral feedback produces the best task experience, stronger emotional-support experience, and more positive immediate self-reported affective changes.

C. Participants and Ethics

We recruited 25 student participants from a university population, including 6 undergraduate students, 13 master's students, and 6 PhD students. The participants had a mean age of 23.4 years (range: 18–29), including 10 women and 15

men. None of the participants had prior experience with the proposed system.

Before the study, participants provided informed consent and were informed about the procedure, four-day schedule, and collected data. They were told that the system was for non-clinical affective art interaction research and emotional-support experience evaluation, not diagnosis, psychotherapy, or medical intervention. Questionnaire responses, speech/text records, facial video, game logs, and ratings were anonymized for research analysis.

Privacy scope. Speech/text, facial video, canvas/action logs, and questionnaires were treated as sensitive affective data under consent and anonymized analysis. Facial and canvas signals were contextual and non-diagnostic. The study does not establish on-device processing or a formal privacy guarantee; future deployments require access control, encryption, retention/deletion and withdrawal procedures, and de-identified release. Additional safeguards are discussed in Supplementary Material, Sec. S4.

D. Experimental Design and Procedure

We conducted a four-day within-subject experiment. Each participant experienced one condition per day and completed all four conditions across separate sessions, reducing carry-over, fatigue, and adaptation effects. Each condition used its own pre- and post-session measurements.

The study used video-based affect sensing and avatar rigging as two factors. Sensing used either video-only S4D (*weak*) or our perception module (*strong*); rigging used either UniRig (*weak*) or our rigging module (*strong*), forming the four conditions in Table V. The cyclic orders were C1–C2–C3–C4, C2–C3–C4–C1, C3–C4–C1–C2, and C4–C1–C2–C3, assigned 6/6/6/7. Sessions for each participant occurred at approximately the same time of day. Separate days, counterbalancing, and daily pre/post measures reduced immediate carry-over, order/adaptation, and baseline-mood effects; full controls are detailed in Supplementary Material, Sec. S4.

At the beginning of each daily session, the participant completed pre-condition affective questionnaires. The participant

TABLE VI
SHORT-TERM AFFECTIVE OUTCOMES ACROSS FOUR CONDITIONS ($M \pm SD$).

Condition	PANAS-PA			PANAS-NA			PSS-10		
	Pre	Post	Change	Pre	Post	Change	Pre	Post	Change
C1	25.96 $\pm$ 4.95	27.84 $\pm$ 5.15	+1.88 $\pm$ 0.97	25.88 $\pm$ 3.73	23.80 $\pm$ 3.49	-2.08 $\pm$ 0.95	19.48 $\pm$ 3.68	17.72 $\pm$ 4.06	-1.76 $\pm$ 0.93
C2	26.60 $\pm$ 4.66	29.64 $\pm$ 4.69	+3.04 $\pm$ 1.10	25.68 $\pm$ 4.46	22.00 $\pm$ 4.28	-3.68 $\pm$ 0.94	18.80 $\pm$ 3.17	15.80 $\pm$ 3.56	-3.00 $\pm$ 0.82
C3	25.56 $\pm$ 5.03	29.16 $\pm$ 4.98	+3.60 $\pm$ 0.82	24.96 $\pm$ 3.74	20.96 $\pm$ 3.93	-4.00 $\pm$ 1.19	18.92 $\pm$ 3.07	15.64 $\pm$ 3.49	-3.28 $\pm$ 0.98
C4	25.88 $\pm$ 3.92	31.16 $\pm$ 3.92	+5.28 $\pm$ 0.84	24.48 $\pm$ 4.78	19.04 $\pm$ 5.06	-5.44 $\pm$ 1.00	18.28 $\pm$ 2.92	13.88 $\pm$ 3.10	-4.40 $\pm$ 0.65

then interacted with the assigned system condition through the Affective-Art Composition Game and the 3D avatar interface. Each interaction followed the same task flow: an initial affective prompt, free geometric composition for emotion externalization, system sensing and state analysis based on the available modalities, 3D avatar feedback, optional avatar-guided creative suggestions, conscious reconstruction, and a short reflective summary. After the interaction, the participant completed post-condition affective questionnaires and subjective experience ratings. Each session lasted approximately 8–10 minutes.

E. Measures

The dependent variables include immediate self-reported affective changes and post-condition subjective experience. Since each condition was conducted on a separate day, affective change was computed within each condition using the corresponding pre- and post-condition scores:

$$\Delta y_{i,c} = y_{i,c}^{post} - y_{i,c}^{pre}, \quad (22)$$

where i denotes participant and c denotes condition. Positive PANAS-PA change indicates increased positive affect, negative PANAS-NA change indicates reduced negative affect, and negative PSS-10 change indicates reduced session-level perceived stress. We used an adapted session-level version of the PSS-10 items to measure perceived stress associated with the current interaction session, rather than long-term perceived stress.

Participants completed PANAS-PA, PANAS-NA, and PSS-10 before and after each session. The Chinese 1–5 post-session questionnaire contained four Empathy, three Social Presence and Affinity, and four Task Experience items. Representative English translations are “The agent accurately captured the emotion I expressed through video,” “I felt that I was interacting with a living entity rather than a machine,” and “I could concretize and express my emotions through colors and shapes.” Dimension scores were averaged; all 11 original Chinese items and translations appear in Supplementary Material, Sec. S4.

The instruments were closed-form and contained no comment field, and no qualitative interview transcript was available; therefore, no verbatim participant comments are reported.

F. Statistical Results Analysis

Table VI shows the largest immediate changes under C4: PA increased by 5.28 ± 0.84 , NA decreased by 5.44 ± 1.00 , and PSS decreased by 4.40 ± 0.65 . Figure 5 shows that C4

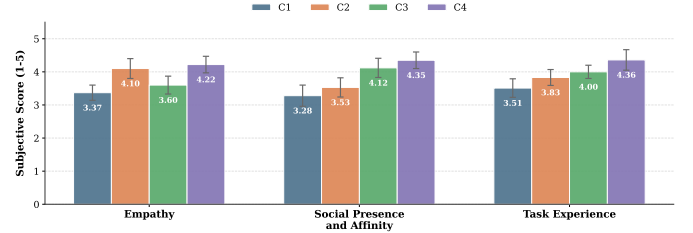


Fig. 5. Subjective outcomes for the four-item Empathy, three-item Social Presence and Affinity, and four-item Task Experience scales. Error bars are standard deviations.

also had the highest Empathy (4.22 ± 0.25), Social Presence and Affinity (4.35 ± 0.25), and Task Experience (4.36 ± 0.31).

Table VII shows strong effects of sensing on Empathy, $F(1, 24) = 218.70$, $p < .001$, $\eta_p^2 = .90$, and rigging on Social Presence and Affinity, $F(1, 24) = 447.86$, $p < .001$, $\eta_p^2 = .95$. Both factors affected Task Experience and all change measures ($p < .001$), with non-significant interactions indicating mostly additive benefits.

G. Analysis of Affective-Art Composition Game

To isolate the role of the Affective-Art Composition Game, we conducted a within-subject with/without-game analysis with the same 25 participants. The existing C4 condition served as the with-game condition, while the without-game condition kept the same settings but removed the composition canvas, game-state input, and game-guidance feedback. The without-game condition was conducted as an additional comparison session after the main four-day study.

Figure 6 shows a representative case. The initial composition was dense, high-contrast, sharp, and heterogeneous; after agent guidance to reduce crowding, soften sharp regions, and create blank space, the reconstruction became more spacious, smoother, and lower in contrast. An illustrative walkthrough connects the prompt, composition, state summary, coordinated avatar response, and reconstruction in Supplementary Material, Sec. S2. It explains the workflow but is not a verbatim participant transcript. Because no open-ended responses or qualitative transcripts were collected, Figure 6 is visual-process evidence rather than participant testimony. Evidence about self-expression is limited to the questionnaire and session-level measures; future work should add open-ended prompts and semistructured interviews. As shown in Table VIII, the with-game condition led to greater improvements in PA, NA, and PSS than the without-game condition. Paired-sample t -tests on the same 25 participants confirmed significant differences

TABLE VII
REPEATED-MEASURES ANOVA FOR THE 2×2 USER-STUDY DESIGN.

Outcome	Effect	F	p	η_p^2	Outcome	Effect	F	p	η_p^2
Empathy	Sensing	218.70	< .001	.90	PA change	Sensing	71.84	< .001	.75
	Rigging	13.36	.001	.36		Rigging	177.66	< .001	.88
	Interaction	2.07	.163	.08		Interaction	1.97	.173	.08
Social Presence and Affinity	Sensing	18.23	< .001	.43	NA change	Sensing	85.36	< .001	.78
	Rigging	447.86	< .001	.95		Rigging	62.77	< .001	.72
	Interaction	0.07	.788	.00		Interaction	0.18	.679	.01
Task Experience	Sensing	43.08	< .001	.64	PSS change	Sensing	55.92	< .001	.70
	Rigging	125.10	< .001	.84		Rigging	122.27	< .001	.84
	Interaction	0.20	.656	.01		Interaction	0.19	.671	.01

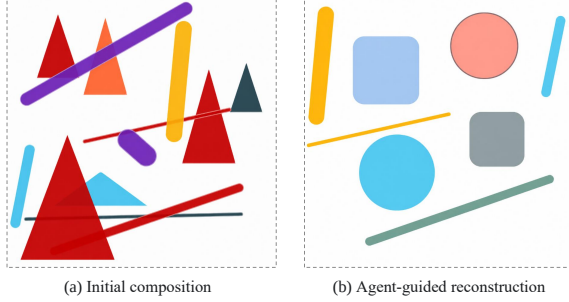


Fig. 6. Representative visual-process case of agent-guided affective art reconstruction. The initial composition shows dense visual organization, strong color contrast, heterogeneous shapes, and sharp contours. After agent-guided reconstruction, the composition becomes more spacious, smoother in shape, and lower in visual contrast.

TABLE VIII
AFFECTIVE CHANGES WITH AND WITHOUT THE AFFECTIVE-ART COMPOSITION GAME.

Condition	ΔPA	ΔNA	ΔPSS
With Game	$+5.28 \pm 0.84$	-5.44 ± 1.00	-4.40 ± 0.65
Without Game	$+3.20 \pm 1.12$	-3.48 ± 0.87	-2.96 ± 0.84
$t(24)$	7.69	-7.50	-6.22
p	< .001	< .001	< .001

across all three measures ($p < .001$). These results show greater immediate self-reported affective change with the game than without it and support its role as a visual mechanism for emotion externalization; they do not establish sustained emotion regulation.

VII. CONCLUSIONS AND LIMITATIONS

This paper presented a 3D embodied agent-mediated affective art system integrating a composition game, semantic-guided video perception, structure-aware avatar rigging, and coordinated responses. Technical experiments improved MAFW sensing and avatar rigging, while the user study associated stronger sensing and embodiment with better perceived empathy, social presence, and immediate self-reported affective changes. These are short-term perceived benefits, not sustained emotion regulation.

The system is non-clinical and does not provide diagnosis, psychotherapy, crisis counseling, or medical intervention; its vocal and behavior libraries also remain limited. Brief sessions

and immediate self-reports do not establish long-term regulation, sustained well-being, or clinical efficacy; longitudinal follow-up is required. Counterbalancing and matched times reduce but cannot eliminate daily mood, time-of-day, or adaptation effects; longer randomized studies should record these covariates. Avatar evidence is limited to successfully rigged humanoids, with no runtime recovery or default-avatar fallback. Extreme geometry and topology can still cause rigging or motion failures; broader benchmarks, validation, failure detection, and recovery are needed (Supplementary Material, Sec. S3).

VIII. ACKNOWLEDGMENT

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REFERENCES

- [1] S. Liu *et al.*, "Towards emotional support dialog systems," in *ACL-IJCNLP*, 2021, pp. 3469–3483.
- [2] Q. Tu, Y. Li, J. Cui, B. Wang, J.-R. Wen, and R. Yan, "Misc: A mixed strategy-aware model integrating comet for emotional support conversation," in *ACL*, 2022, pp. 308–319.
- [3] J. Zhou, Z. Chen, B. Wang, and M. Huang, "Facilitating multi-turn emotional support conversation with positive emotion elicitation: A reinforcement learning approach," in *ACL*, 2023, pp. 1714–1729.
- [4] C. Zou *et al.*, "Affective flow language model for emotional support conversation," *arXiv preprint arXiv:2602.08826*, 2026.
- [5] C. Hu, Z. Lin, N. Zhang, and L.-J. Ji, "Ai-empowered imagery writing: integrating ai-generated imagery into digital mental health service," *Frontiers in psychiatry*, vol. 15, p. 1434172, 2024.
- [6] X. Luo *et al.*, "Emergence of artificial intelligence art therapies (aiats) in mental health care: a systematic review," *International Journal of Mental Health Nursing*, vol. 33, no. 6, pp. 1743–1760, 2024.
- [7] P. Newland and B. A. Bettencourt, "Effectiveness of mindfulness-based art therapy for symptoms of anxiety, depression, and fatigue: A systematic review and meta-analysis," *Complementary Therapies in Clinical Practice*, vol. 41, p. 101246, 2020.
- [8] B. Han *et al.*, "The effects of visual art therapy on adults with depressive symptoms: A systematic review and meta-analysis," *International journal of mental health nursing*, vol. 33, no. 5, pp. 1183–1196, 2024.
- [9] R. Joschko, C. Klatte, W. A. Grabowska, S. Roll, A. Berghöfer, and S. N. Willich, "Active visual art therapy and health outcomes: A systematic review and meta-analysis," *JAMA network open*, vol. 7, no. 9, p. e2428709, 2024.
- [10] M. Neuville and B. Pfiffner, "Guided drawing: Facilitating trauma narratives for young children," *Journal of Creativity in Mental Health*, vol. 14, no. 4, pp. 519–531, 2019.
- [11] E. C. Ciasca *et al.*, "Art therapy as an adjuvant treatment for depression in elderly women: a randomized controlled trial," *Brazilian Journal of Psychiatry*, vol. 40, pp. 256–263, 2018.

- [12] H. Hattori, C. Hattori, C. Hokao, K. Mizushima, and T. Mase, "Controlled study on the cognitive and psychological effect of coloring and drawing in mild alzheimer's disease patients," *Geriatrics & gerontology international*, vol. 11, no. 4, pp. 431–437, 2011.
- [13] J. Kim and Y. J. Chung, "A case study of group art therapy using digital media for adolescents with intellectual disabilities," *Frontiers in psychiatry*, vol. 14, p. 1172079, 2023.
- [14] X. Du, P. An, J. Leung, A. Li, L. E. Chapman, and J. Zhao, "Deepthink: Designing and probing human-ai co-creation in digital art therapy," *International journal of human-computer studies*, vol. 181, p. 103139, 2024.
- [15] A. Boadi-Agyemang and M. Park, "Designing interactive agents to support emotion regulation in the workplace through guided art-making," in *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 2024, pp. 257–262.
- [16] C. Wu, I. K. Kumar, and R. Picard, "Navigating emotions through art: Recommendations for designing art-therapy based chatbots for trauma-impacted youth," in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–7.
- [17] L. Sun, P. Chen, W. Xiang, P. Chen, W.-y. Gao, and K.-j. Zhang, "Smartpaint: A co-creative drawing system based on generative adversarial networks," *Frontiers of Information Technology & Electronic Engineering*, vol. 20, no. 12, pp. 1644–1656, 2019.
- [18] H. Shen, L. Shen, W. Wu, and K. Zhang, "Ideationweb: Tracking the evolution of design ideas in human-ai co-creation," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–19.
- [19] C. Wang, L. Wu, X. Liu, X. Li, L. Meng, and X. Meng, "Anything to glyph: Artistic font synthesis via text-to-image diffusion model," in *SIGGRAPH Asia 2023 Conference Papers*. ACM, 2023, pp. 1–11.
- [20] X. Li, L. Meng, L. Wu, M. Li, and X. Meng, "DreamFont3D: Personalized text-to-3d artistic font generation," in *ACM SIGGRAPH 2024 Conference Papers*. ACM, 2024, pp. 1–11.
- [21] X. Jiang *et al.*, "Dflew: A large-scale database for recognizing dynamic facial expressions in the wild," in *Proceedings of the 28th ACM international conference on multimedia*, 2020, pp. 2881–2889.
- [22] Y. Wang *et al.*, "Ferv39k: A large-scale multi-scene dataset for facial expression recognition in videos," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 20922–20931.
- [23] Y. Liu *et al.*, "Mafw: A large-scale, multi-modal, compound affective database for dynamic facial expression recognition in the wild," in *Proceedings of the 30th ACM international conference on multimedia*, 2022, pp. 24–32.
- [24] Z. Lian *et al.*, "Affectgpt: A new dataset, model, and benchmark for emotion understanding with multimodal large language models," *arXiv preprint arXiv:2501.16566*, 2025.
- [25] M. Jung, Y. Lim, S. Kim, J. Y. Jang, S. Shin, and K.-H. Lee, "An emotion-based korean multimodal empathetic dialogue system," in *Proceedings of the Second Workshop on When Creative AI Meets Conversational AI*. Association for Computational Linguistics, 2022, pp. 16–22.
- [26] J. Hu, Y. Huang, X. Hu, and Y. Xu, "The acoustically emotion-aware conversational agent with speech emotion recognition and empathetic responses," *IEEE Transactions on Affective Computing*, vol. 14, no. 1, pp. 17–30, 2023.
- [27] R. Sanjeeva, R. Iyer, P. Apputhurai, N. Wickramasinghe, and D. Meyer, "Perception of empathy in mental health care through voice-based conversational agent prototypes: experimental study," *JMIR Formative Research*, vol. 9, no. 1, p. e69329, 2025.
- [28] H. Fei, H. Zhang, B. Wang, L. Liao, Q. Liu, and E. Cambria, "EmpathyEar: An open-source avatar multimodal empathetic chatbot," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*. Association for Computational Linguistics, 2024, pp. 61–71.
- [29] J. Mercado, I. E. Espinosa-Curiel, and J. Martínez-Miranda, "Embodied conversational agents providing motivational interviewing to improve health-related behaviors: Scoping review," *Journal of medical Internet research*, vol. 25, p. e52097, 2023.
- [30] R. Sanjeeva, R. Iyer, P. Apputhurai, N. Wickramasinghe, and D. Meyer, "Empathic conversational agent platform designs and their evaluation in the context of mental health: systematic review," *JMIR Mental Health*, vol. 11, p. e58974, 2024.
- [31] F.-C. Yang, P. Acevedo, S. Guo, M. Choi, and C. Mousas, "Embodied conversational agents in extended reality: A systematic review," *IEEE Access*, vol. 13, pp. 84 833–84 854, 2025.
- [32] L. Trautmann, "Emotions evoked by geometric patterns," *J*, vol. 4, no. 3, pp. 376–393, 2021.
- [33] M. Bar and M. Neta, "Visual elements of subjective preference modulate amygdala activation," *Neuropsychologia*, vol. 45, no. 10, pp. 2191–2200, 2007.
- [34] M. Rankanen, M. Leinikka, C. Groth, P. Seitamaa-Hakkarainen, M. Mäkelä, and M. Huotilainen, "Physiological measurements and emotional experiences of drawing and clay forming," *The Arts in Psychotherapy*, vol. 79, p. 101899, 2022.
- [35] N. Colnerić and J. Demšar, "Emotion recognition on twitter: Comparative study and training a unison model," *IEEE transactions on affective computing*, vol. 11, no. 3, pp. 433–446, 2020.
- [36] J. Deng and F. Ren, "A survey of textual emotion recognition and its challenges," *IEEE Transactions on Affective Computing*, vol. 14, no. 1, pp. 49–67, 2023.
- [37] X. Qian, S. Wu, J. Chen, W. Shen, and J. Yu, "Multi-party conversation modeling for emotion recognition," *IEEE Transactions on Affective Computing*, vol. 15, no. 3, pp. 751–768, 2024.
- [38] W.-C. Lin and C. Busso, "Chunk-level speech emotion recognition: A general framework of sequence-to-one dynamic temporal modeling," *IEEE Transactions on Affective Computing*, vol. 14, no. 2, pp. 1215–1227, 2023.
- [39] W. Chen, X. Xing, P. Chen, and X. Xu, "Vesper: A compact and effective pretrained model for speech emotion recognition," *IEEE Transactions on Affective Computing*, vol. 15, no. 3, pp. 1711–1724, 2024.
- [40] H. Lian, C. Lu, Y. Zhao, S. Li, T. Qi, and Y. Zong, "Exploring corpus-invariant emotional acoustic feature for cross-corpus speech emotion recognition," *Expert Systems with Applications*, vol. 258, p. 125162, 2024.
- [41] C. Lu *et al.*, "Hierarchical distribution adaptation for unsupervised cross-corpus speech emotion recognition," in *Interspeech 2024*. ISCA, 2024, pp. 3739–3743.
- [42] S. Li, C. Lu, Y. Zhao, H. Lian, T. Qi, and Y. Zong, "Low-rank joint distribution adaptation for cross-corpus speech emotion recognition," *Knowledge-Based Systems*, vol. 315, p. 113260, 2025.
- [43] D. Huang, D. Jiang, S. Zhou, L. Chen, J. Lin, and E. Cambria, "Emotional lateralization-inspired spatiotemporal neural networks for EEG-based emotion recognition," *Information Fusion*, vol. 129, p. 104053, 2026.
- [44] D. Huang, S. Zhou, and D. Jiang, "Generator-based domain adaptation method with knowledge free for cross-subject EEG emotion recognition," *Cognitive Computation*, vol. 14, no. 4, pp. 1316–1327, 2022.
- [45] S. Zhou, D. Huang, C. Liu, and D. Jiang, "Objectivity meets subjectivity: A subjective and objective feature fused neural network for emotion recognition," *Applied Soft Computing*, vol. 122, p. 108889, 2022.
- [46] J. Z. Wang *et al.*, "Unlocking the emotional world of visual media: An overview of the science, research, and impact of understanding emotion," *Proceedings of the IEEE*, vol. 111, no. 10, pp. 1236–1286, 2023.
- [47] S. Li and W. Deng, "Deep facial expression recognition: A survey," *IEEE transactions on affective computing*, vol. 13, no. 3, pp. 1195–1215, 2022.
- [48] W. Chen, D. Zhang, M. Li, and D.-J. Lee, "Scam: Spatial-temporal and channel attention module for dynamic facial expression recognition," *IEEE Transactions on Affective Computing*, vol. 14, no. 1, pp. 800–810, 2023.
- [49] Z. Tong, Y. Song, J. Wang, and L. Wang, "Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 10 078–10 093.
- [50] H. Li, H. Niu, Z. Zhu, and F. Zhao, "Intensity-aware loss for dynamic facial expression recognition in the wild," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, no. 1, 2023, pp. 67–75.
- [51] L. Sun, Z. Lian, B. Liu, and J. Tao, "Mae-dfer: Efficient masked autoencoder for self-supervised dynamic facial expression recognition," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 6110–6121.
- [52] —, "Hicmae: Hierarchical contrastive masked autoencoder for self-supervised audio-visual emotion recognition," *Information Fusion*, vol. 108, p. 102382, 2024.
- [53] H. Chen, H. Huang, J. Dong, M. Zheng, and D. Shao, "Finecliper: Multimodal fine-grained clip for dynamic facial expression recognition with adapters," in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 2301–2310.
- [54] Y. Chen *et al.*, "Static for dynamic: Towards a deeper understanding of dynamic facial expressions using static expression data," *IEEE Transactions on Affective Computing*, vol. 17, no. 1, pp. 438–451, 2026.
- [55] S. Yang, H.-M. Sun, S. Feng, and R.-S. Jia, "Self-supervised dynamic facial expression recognition with efficient encoding and redundant tokens compression," *Neurocomputing*, vol. 668, p. 132274, 2026.
- [56] J.-H. Hsu and C.-H. Wu, "Applying segment-level attention on bi-modal transformer encoder for audio-visual emotion recognition," *IEEE Transactions on Affective Computing*, vol. 14, no. 4, pp. 3231–3243, 2023.
- [57] Y. Wang, Z. Qi, X. Li, J. Liu, X. Meng, and L. Meng, "Multi-channel attentive weighting of visual frames for multimodal video classification," in *2023 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2023, pp. 1–8.
- [58] Y. Wang, L. Meng, H. Ma, Y. Wang, H. Huang, and X. Meng, "Modeling event-level causal representation for video classification," in *Proceedings of the 32nd ACM International Conference on Multimedia*. ACM, 2024, pp. 3936–3944.
- [59] S. Zhao *et al.*, "Sdrs: Sentiment-aware disentangled representation shifting for multimodal sentiment analysis," *IEEE Transactions on Affective Computing*, vol. 16, no. 3, pp. 1802–1813, 2025.
- [60] D. Wu, D. Yang, S. Zhao, C. Ma, and Y. Zhou, "An empirical study on configuring in-context learning demonstrations for unleashing LLMs' sentimental perception capability," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 67 986–67 999.
- [61] W. Guo, Y. Zhang, X. Cai, L. Meng, J. Yang, and X. Yuan, "LD-MAN: Layout-driven multimodal attention network for online news sentiment recognition," *IEEE Transactions on Multimedia*, vol. 23, pp. 1785–1798, 2021.

- [62] S. Kim, H. Kim, H. Do, and G. Lee, "Multimodal cognitive reframing therapy via multi-hop psychotherapeutic reasoning," in *NAACL*, 2025, pp. 4863–4880.
- [63] S. Kim, H. Kim, J. Lee, Y. Jeon, and G. Lee, "Mirror: Multimodal cognitive reframing therapy for rolling with resistance," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 14 851–14 880.
- [64] Y. Chen *et al.*, "Soulchat: Improving llms' empathy, listening, and comfort abilities through fine-tuning with multi-turn empathy conversations," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 1170–1183.
- [65] Z. Chen, T. Qi, C. Lu, and W. Zheng, "DisenEmo: Learning disentangled emotional representation from facial motion for 3D talking head generation," in *2025 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2025, pp. 289–294.
- [66] P. Ekman and W. V. Friesen, *Facial Action Coding System: A Technique for the Measurement of Facial Movement*, Consulting Psychologists Press, Palo Alto, CA, USA, 1978.
- [67] T. Ren *et al.*, "Grounded sam: Assembling open-world models for diverse visual tasks," *arXiv preprint arXiv:2401.14159*, 2024.
- [68] N. Ravi *et al.*, "Sam 2: Segment anything in images and videos," in *The Thirteenth International Conference on Learning Representations*, 2025.
- [69] Z. Zhao *et al.*, "Hunyuan3d 2.0: Scaling diffusion models for high resolution textured 3d assets generation," *arXiv preprint arXiv:2501.12202*, 2025.
- [70] J.-P. Zhang, C.-F. Pu, M.-H. Guo, Y.-P. Cao, and S.-M. Hu, "One model to rig them all: Diverse skeleton rigging with UniRig," *ACM Transactions on Graphics*, vol. 44, no. 4, pp. 1–18, 2025.
- [71] Z. Gao *et al.*, "Funasr: A fundamental end-to-end speech recognition toolkit," in *Interspeech 2023*. ISCA, 2023, pp. 1593–1597.
- [72] Z. Du *et al.*, "Cosyvoice: A scalable multilingual zero-shot text-to-speech synthesizer based on supervised semantic tokens," *arXiv preprint arXiv:2407.05407*, 2024.
- [73] T. Shen, J. Wang, X. Zhang, and E. Cambria, "Reasoning with trees: Faithful question answering over knowledge graph," in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 3138–3157.
- [74] J. S. Chung, A. Nagrani, and A. Zisserman, "Voxceleb2: Deep speaker recognition," in *Interspeech 2018*. ISCA, 2018, pp. 1086–1090.
- [75] A. Mollahosseini, B. Hasani, and M. H. Mahoor, "Affectnet: A database for facial expression, valence, and arousal computing in the wild," *IEEE transactions on affective computing*, vol. 10, no. 1, pp. 18–31, 2019.
- [76] S. Blackman, "Rigging with mixamo," in *Unity for Absolute Beginners*. Springer, 2014, pp. 565–573.
- [77] N. Isozaki, S. Ishima, Y. Yamada, Y. Obuchi, R. Sato, and N. Shimizu, "Vvoid studio: a tool for making anime-like 3d characters using your imagination," in *SIGGRAPH Asia 2021 Real-Time Live!*, 2021, pp. 1–1.
- [78] Z. Xu, Y. Zhou, E. Kalogerakis, C. Landreth, and K. Singh, "Rignet: Neural rigging for articulated characters," *ACM Transactions on Graphics*, vol. 39, no. 4, pp. 58:1–58:14, 2020.
- [79] X. Zhang *et al.*, "A distributed architecture digital human service system powered by large language models," in *Neural Information Processing: 31st International Conference, ICONIP 2024, Proceedings, Part VI*, ser. Communications in Computer and Information Science, vol. 2287. Springer, 2025, pp. 283–292.



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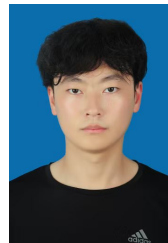
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